3 METHODOLOGY

Drawing from cognitive science research (Maier, 1996; Uttal et al., 2013), we organize spatial reasoning into two key dimensions: **Intrinsic vs. Extrinsic** and **Static vs. Dynamic**. The first dimension differentiates between **Intrinsic vs. Extrinsic** information. Intrinsic information refers to the essential characteristics and relationships that define an object. Extrinsic information refers to the relation among objects in a group, relative to one another or to an overall framework. The second dimension, **Static vs. Dynamic**, centers on movement. Movement can alter intrinsic information, such as through folding, cutting, or rotation. It can also shift an object's position relative to other objects and the surrounding environment.

This framework comprehensively covers existing task classifications by placing them into four distinct quadrants. This creates a 2x2 taxonomy that categorizes spatial reasoning into four distinct quadrants: **Intrinsic-Static (I-S)** tasks involve analyzing the internal properties of a single, unchanged object; **Extrinsic-Static (E-S)** tasks assess the relationships between multiple objects in a fixed scene; **Intrinsic-Dynamic (I-D)** tasks require mentally simulating transformations on a single object; and **Extrinsic-Dynamic (E-D)** tasks involve reasoning about the changing spatial relationships between multiple objects.

3.1 TASKS DESIGN

We designed 10 cognitive science-based tasks to probe spatial reasoning. Figure 2 provides a visual guide to this categorization, showing how various spatial tasks map onto our **Spatial-DISE** taxonomy. The 10 tasks we designed not only fully map to the four quadrants of the DISE framework, but their design inspiration also stems from classical psychometric tests. These tasks are specifically designed to assess core spatial abilities such as mental rotation and spatial visualization (see Appendix A.1 for detailed correspondence).

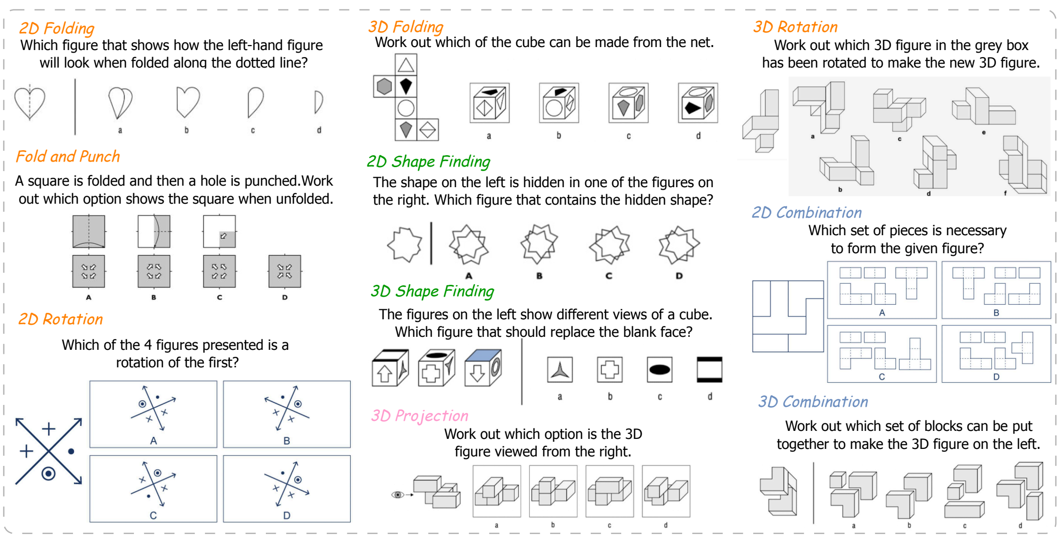

Figure 2: 10 Tasks in Spatial-DISE Bench. Orange shows the Intrinsic-Dynamic Tasks, Green shows the Intrinsic-Static Tasks, Pink shows the Extrinsic-Static Tasks and Blue shows the Extrinsic-Dynamic Tasks.

Intrinsic-Static Tasks. These tasks evaluate the understanding of an object's fixed, internal spatial properties without transformation. This is assessed through **2D/3D Shape Finding**, which requires identifying a hidden shape within a more complex figure or determining a cube's missing face from other views, thereby testing the static analysis of intrinsic part-whole relationships.

Intrinsic-Dynamic Tasks. These tasks test the ability to mentally manipulate the internal properties of an object, requiring pure mental simulation. This includes **2D/3D Rotation**, a classic test of mental transformation that requires predicting an object's appearance after rotation, and **2D/3D Folding & Fold&Punch**, which tests the outcome of folding a 2D net into a 3D shape or unfolding a punched paper.

Extrinsic-Static Tasks. These tasks investigate the understanding of fixed spatial relationships from an external viewpoint. This is probed using **3D Projection**, which requires identifying the correct 2D orthographic projection of a 3D object from a specified external direction, a task dependent on the extrinsic relationship between observer and object.

Extrinsic-Dynamic Tasks. These tasks assess the ability to reason about the changing relationships between multiple objects or parts. The primary tasks here are **2D/3D Combination**, which require mentally assembling separate parts into a coherent whole and thus tests the ability to simulate how components must move, orient, and connect.

3.2 BENCHMARK CURATION

To ensure the scientific rigor and validity of the dataset, a 3-stage curation pipeline was employed. This process integrates real-world data from open sources with a scalable synthetic data generation, followed by human quality control.

Stage 1: Real-world Data Collection. The initial phase aimed to establish a conceptual foundation and a repository of templates for subsequent data synthesis. We collected a corpus of existing, high-quality spatial reasoning problems from publicly available and validated sources, including academic psychometric tests and professional aptitude assessments. This phase yielded an initial corpus of 1180 VQA pairs, providing a diverse set of concepts and structures that informed the automated generation process. Detailed real-world data collection is presented in Appendix A.3.

Stage 2: Scalable Synthetic Data Generation. As illustrated in Figure 1 d, this core stage was designed to overcome the scale limitations of existing benchmarks, particularly for dynamic and 3D tasks. Leveraging the Blender engine, we transformed the concepts from the initial corpus into a scalable, automated pipeline. This pipeline follows a general paradigm, customized for each of the five 3D task types: **1) Initialization and Seeding:** Each generation task begins with a unique $question\_id$, which is hashed to create a reproducible random seed, ensuring the uniqueness and verifiability of every generated instance. **2) Core Asset Generation:** We generate the core 3D object for a given problem. This includes creating complex, irregular shapes for tasks like 3D Rotation or generating cubes with unique face textures for 3D Folding and 3D Shape Finding. **3) Question and Correct Answer Rendering:** We render the question and correct answer images from optimal camera perspectives. **4) Systematic Distractor Generation**: To ensure the diagnostic challenge of each item, the pipeline implements a suite of tiered strategies to create plausible, near-miss distractors. These strategies include: - *Geometric Variations:* Introducing subtle alterations to the core object's geometry, such as adding or removing components. - *Pattern/Orientation Errors:* Generating incorrect texture layouts or orientations on the faces of an object. - *Incorrect Views:* Rendering a correct object from an incorrect orthographic perspective for projection tasks. - *Component Replacement:* Swapping a correct part with a geometrically similar but incorrect one in assembly tasks. **5) Controlled Rendering and Formatting:** All question, correct answer, and distractors are rendered in a controlled virtual environment with consistent lighting, materials, and camera parameters. The final output is a standardized VQA data pair. Detailed illustration and pseudocode of synthetic data generation is shown in Appendix A.4.

Stage 3: Rigorous Human Quality Control. Following generation, all benchmark synthetic instances underwent a rigorous manual verification process to guarantee the benchmark's integrity and reliability. The review protocol assessed each instance against three quality criteria. **1) Solution Uniqueness**: Each problem must have a single, unambiguous correct answer. **2) Accuracy and Clarity**: All images must be free of rendering artifacts, and the corresponding questions must be clearly articulated. All options must be valid according to the task's criteria. **3) Redundancy Elimination**: The instance should not be logically or visually redundant with other items in the dataset. Instances failing to meet these standards were removed from the final dataset. Combined with real-world data, two sets of Spatial-DISE are created:

- **Spatial-DISE Bench**: An evaluation set of 559 carefully selected VQA pairs, covering all 10 task types and four DISE dimensions, designed for model benchmarking.
- **Spatial-DISE 12K**: A large-scale dataset consisting of over 12,000 verifiable VQA pairs cover five 3D tasks, intended as a valuable resource for the future training and fine-tuning of spatial reasoning capabilities in VLMs.

4 EVALUATION ON SPATIAL-DISE BENCH

4.1 EXPERIMENT SETTING

Benchmark Models. For the evaluation, we select a diverse set of **32** models across **10** model families. Our selection includes both proprietary and open-source models, spanning general foundation models, reasoning models, and spatial-specified models. For proprietary foundation models, we evaluated Claude3.7-Sonnet, Doubao1.5-VL (Guo et al., 2025b), Gemini-2.0-Flash, Gemini-2.5-Flash (w/ and w/o thinking), GPT-4.1-nano, GPT-4o, GPT-4o-mini, GPT-5, and o4-mini. For open-source foundation models, we evaluated InternVL-3-[8B/14B/38B] (Zhu et al., 2025), Llama-3.2-11B-Vision (Grattafiori et al., 2024), Kimi-VL-A3B (Du et al., 2025), Ovis2-[8B/16B] (Lu et al., 2024), Cambrian-[8B/13B] (Tong et al., 2024) and Qwen2.5-VL-[3B/7B/32B] (Bai et al., 2025). For reasoning models, we evaluated LLaVA-CoT (Xu et al., 2024), LMM-R1 (Peng et al., 2025), VLM-R1 (Shen et al., 2025), VLAA-Thinker-[3B/7B] (Chen et al., 2025), Kimi-VL-A3B-Thinking (Du et al., 2025) and Doubao-1.5-thinking (Guo et al., 2025b). For spatial-specified models, we evaluate SpaceThinker (Chen et al., 2024), SpaceOM (Chen et al., 2024) and SpaceR (Ouyang et al., 2025).

Baseline. For comparison, we include Random Guessing and Human Performance. To establish a robust, psychometrically sound human baseline, we recruited 54 diverse participants (ages 15-55) and employed a matrix-sampling design to collect 1,679 valid responses. This design ensured each question was answered by an average of three unique participants. The final human performance is reported as the average accuracy across all responses and cross-validated with Item Response Theory (IRT). More details of human performance in Appendix B.1.

Implementation Details. We evaluate multiple-choice accuracy using exact match via the VLMEvalKit (Duan et al., 2025). Deepseek-R1 (Guo et al., 2025a) is used to parse answers from malformed model outputs. Additional implementation details are provided in the Appendix B.2.

4.2 MAIN RESULTS

Our comprehensive evaluation reveals that spatial reasoning remains a significant and universal challenge for current VLMs. Table 2, 3 present the main results of our evaluation. More results are presented in Appendix B.3. We summarize the key findings as followed:

Spatial reasoning remains a universal challenge. The overall performance across all 32 tested models was low, with average accuracy of 28.4%, only marginally above random chance (25%) and falling drastically short of the human baseline (76.8%). Of all models evaluated, the reasoning-enhanced Doubao1.5-VL-thinking achieved the highest overall accuracy at 42.0%. This widespread underperformance indicates a critical weakness in tasks requiring genuine mental transformation, highlighting a failure to move beyond pattern recognition to true spatial cognition.

Multi-Step transformations overwhelm VLMs reasoning. Models demonstrate a particular vulnerability to tasks requiring a sequence of mental transformations. The Fold and Punch task, which requires simulating a fold, a punch, and then an unfold, serves as a clear example of this failure. Even the top-performing model, Doubao-1.5-thinking, only achieved 30.8% accuracy, while the average of all models is only 25.4%, performed near random chance. This indicates that while a model might handle a single transformation, its ability to maintain a coherent mental state breaks down across multiple steps. This suggests a critical deficit in "spatial working memory," preventing models from reliably tracking an object through a sequence of changes.

Post-training shows improvement but not enough. The results reveal that post-training with reinforcement learning or fine-tuning on spatial datasets offers limited improvements. While models like Doubao-1.5-thinking and SpaceThinker showed performance gains, their absolute accuracies remain low and far from the human baseline.

Static comprehension is not a solved precursor to dynamic reasoning. Counter-intuitively, the results show that proficiency in static reasoning is not a prerequisite for dynamic reasoning. Several top models perform better on dynamic tasks than static ones. For example, Gemini2.0-Flash scored significantly higher on dynamic tasks (38.3%) than on static tasks (23.6%). Doubao-1.5-thinking even outperform human performance in Extrinsic-Dynamic questions. This suggests that models are